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Submitted on: 30/5/26
Quiz 1

Given the following binary numbers which are in 2's complement

(a) 0110010

(b)1011100

(i) What is the decimal equivalent of the above data (a) and data(b), also indicate whether the number is a positive or a negative number

q1 (a)

$(-)64_{10}$	32_{10}	16_{10}	8_{10}	4_{10}	2_{10}	1_{10}
0_2	1_2	1_2	0_2	0_2	1_2	0_2

$0110010_2 = 2+16+32$

$= (+)50_{10}$

This is a positive number because the MSB of the binary number on the farthest left hand-side is a 0.

q2) (b)

$(-)64_{10}$	32_{10}	16_{10}	8_{10}	4_{10}	2_{10}	1_{10}
1_2	0_2	1_2	1_2	1_2	0_2	0_2

$1011100_2 = 4+8+16+ (-64)$

$= (-)36_{10}$

This is a negative number because the MSB of the binary number is a 1, giving it a negative value.

(ii) Perform addition of these two 2's complement numbers and from the result, indicates if this is positive or negative number? Also find the decimal equivalent of the result obtained.

1	1	1					
	0	1	1	0	0	1	0
+	1	0	1	1	1	0	0
1	0	0	0	1	1	1	0

The answer comes out to $1\ 0\ 0\ 0\ 1\ 1\ 1\ 0_2$ but we will discard the further left binary digit, making it $0\ 0\ 0\ 1\ 1\ 1\ 0_2$. This number is no longer negative because its MSB is a 0 instead of a 1.

The decimal equivalent of this would be $50_{10} + (-36_{10}) = 14_{10}$